SDC HACKATHON 2026 · DOMAIN 2 · ORBYTE SMART SME CHALLENGE

KiraOrder

AT A GLANCE

Five-stage loop, three roles, two tenants, one audit chain — built in four days on synthetic data only.

It knows when not to guess — and who to ask.

THE PRODUCT, IN ONE SENTENCE

Turns a mixed-language WhatsApp order into an approved delivery order — and stops exactly where a guess would cost money.

WHO IT IS FOR

Micro and small Malaysian food wholesalers. One clerk takes the evening orders, one owner approves the exceptions. No ERP, three languages per message.

TEAM

Falcons · 5 members · Xiamen University Malaysia

SDG

8 Decent Work and Economic Growth · 8.2, 8.3

STATUS

Working prototype on a public link · synthetic data

6:04 pm. One message. Six lookups, four systems.

THE MESSAGE, AS IT ARRIVES

> “3 ctn Ayam A1, 20kg bwg bsr India,
> 5 包美禄粉 , kalau ada”
>
> “Milo 5 pack — kalau RM95 boleh?”

BASELINE · PLANNING ESTIMATE, NOT MEASURED

≈ 210 s	≤ 30 %	0
median handling time per 5-line order	transcription or stockout discrepancies	independent SME pilots

Team estimate from the brief and domain research. Phase 2 runs the pilot that replaces them.

01	Which Milo — the 1 kg pack or the 2 kg pack?	EXCEL PRICE LIST
02	Which tier price does this customer buy on?	OWNER’S HEAD
03	Which egg grade: A, B or C — same price today?	CATALOGUE
04	Is the onion in stock, and how much of it?	WHITEBOARD
05	Does “kalau ada” mean reduce the line, or drop it?	NOWHERE
06	Does RM 95 sit below her tier price?	CALCULATOR

The bottleneck is not reading the message. It is deciding what must not be guessed.

Source: SDC Domain 2 brief, pp. 3–4 · synthetic order used in the demo

Three answers exist. All three optimise the message.

Each one gets better at parsing. None of them can supply a fact the message does not contain.

THE FOUR FACTS ARE NOT IN THE MESSAGE

- The price file
- The customer’s tier
- The stock board
- The owner’s phone

No reader — human or model — can invent them.

01

By hand

Ask the customer, read the price file, walk to the whiteboard, type the DO.

WHAT IT MISSES

Nothing except time — and the whole thing collapses when eleven messages land at once.

→ **20–40 min per order**

02

A translating bot

Turn Malay, Chinese and English into a clean, structured order list.

WHAT IT MISSES

The order looks perfect and is still wrong. “Kalau ada” and the pack size are not language problems.

→ **Confidence without correctness**

03

A flag-everything checker

Mark suspicious lines red and stop the flow so a human looks.

WHAT IT MISSES

It stops the work without finishing it. A red flag is a question, not an answer.

→ **Alert fatigue, then ignored**

THE BOUNDARY

For a shop selling one product at one price, any of these is enough. Our user has four pack sizes, six tier prices and a stock board on the wall — that is where all three stop.

It stops. Then it knows who to ask.

One message in, four destinations out. The primary button always shows the state of the decision, so nobody has to remember the workflow.

THE SYSTEM

Settles what data settles

Stock, pack size and tier price are read off the catalogue. “Kalau ada” is answered by the count, never by a guess.

No human needed

THE CUSTOMER

Answers the meaning

A ready-made question in her own language: “Boss, Milo nak 1kg atau 2kg?” The clerk copies it — we never send on her behalf.

Drafted, not sent

THE OWNER

Decides the money

Price below tier, order above threshold, or a flagged customer. He sees the reason, the numbers and her exact words.

Two taps, on a phone

THE CLERK

Approves the rest

A routine order from a regular needs no escalation. One button: approve, and the DO exists, stock moves, the log gets a line.

One button, no typing

THE BRIEF’S FIVE STAGES, IN ORDER — MAPPED 1:1

Trigger → pasted order message

Connected data → catalogue, aliases, prices, stock

Decision logic → rules settle what data settles

Human approval → exceptions reach a named role

Visible outcome → DO issued, stock moved, log kept

Every stage that data can settle is automated. Every stage that can move money needs a person.

The link is live. Watch five things.

Synthetic tenant · Malaysian food wholesale · order DJ0741 → delivery order DO-T1-0002 · RM 1,395.00 against an RM 1,500.00 approval threshold

01 **Paste the order**

Three languages, abbreviations, no SKU codes. No template and no field mapping.

02 **Watch the doubt, not the parse**

The unit was never stated and the egg grade was left open. The button asks; it does not pick.

03 **Watch the money leave the clerk**

RM 100 → RM 95 sits below this customer's tier. The owner approves; nothing is auto-accepted.

04 **Watch the record**

Stock moves, DO-T1-0002 exists, reply drafted, and three names sit on one traceable chain.

05 **Watch the guardrail**

The clerk's token calling the owner endpoint returns 403 — enforced in the backend, not in the UI.

IF THE LINK DROPS

Backup recording plus a Local Demo mode: browser-only, synthetic data, no backend and no model call.

WHAT THE DEMO DOES NOT SHOW

- No WhatsApp connection — the clerk pastes.
- No accounting integration — a synthetic field.
- Identity is demo tokens, not single sign-on.
- PDPA handling is designed, not certified.

One order, four layers.

Each layer has one owner, one refusal and one output. That is why the order can be audited line by line.

01

Order understanding

AI · the model

Turns a line into catalogue candidates with verbatim spans.

Refuses: prices, totals, IDs.

→

02

Business verification

DATA · the rules

Pack size and unit come from the catalogue; amounts are exact decimal on the tier.

Refuses: inference.

→

03

Human decision

PEOPLE · the roles

Every doubt carries the thing needed to answer it, and names the role that may release it.

Refuses: auto-approval.

→

04

Business output

SYSTEM · the record

DO issued, stock deducted, reply drafted, one audit chain.

Refuses: silent edits.

BUSINESS PATH	React + TypeScript (GitHub Pages) → FastAPI (Replit) → Neon PostgreSQL 17 · row-level tenant isolation
AI PATH	FastAPI → PII masking → vLLM + Qwen2.5-7B-Instruct-AWQ (RunPod) → closed-list JSON
GUARDRAILS	no model-set prices · spans only · roles enforced in the backend · every action logged · stock locked at approval

AI understands the message. Business rules verify the facts. People make the decisions.

06 · THE PRINCIPLE

AI reads. Data and rules judge. A person decides.

Structural, not politely requested: the model never sees a price, cannot write a total and cannot approve anything.

01 AI READS

Reads. Never decides.

One job: turn the line into a catalogue candidate. Closed list, verbatim spans, no arithmetic.

```
{
  "span": "5 包美禄粉 ",
  "fam": "FAM-MILO-POWDER",
  "qty": 5, "size": null,
  "uom": " 包 "
}
```

NO PRICE · NO TOTAL · NO CUSTOMER ID

02 DATA AND RULES JUDGE

Judges, with the catalogue.

- Pack size, grade and unit are read off the catalogue’s structure, not inferred.
- The amount is exact decimal arithmetic on that customer’s tier — never a model output.
- Stock is re-checked under a row lock at approval, so a promise cannot outrun the shelf.

03 A PERSON DECIDES

Decides, and signs.

- Every DO carries a human name: the clerk for routine, the owner for exceptions.
- The owner sees the reason, the numbers, her exact words and what she changed — two taps on a phone.
- The AI draft is stored beside the human’s final, field by field, with a reason code.

WHY THIS IS NOT A PROMPT

A prompt injection buys at most one wrong line — which a person still approves, and which cannot move a single sen.

Every rule above is enforced in the backend. The interface only reflects it.

Every delivery order carries a human name.

Permissions are enforced in the backend, and the escalation rule is one line of tenant configuration — not a habit the clerk has to remember.

WHO MAY RELEASE THE ORDER	CLERK	OWNER	NOBODY
Routine order under threshold	●	○	○
Price below this customer's tier	○	●	○
Order at or above the tenant threshold (RM 1,500 / RM 800)	○	●	○
Customer carrying an accounting flag	○	●	○
A blocking doubt still unresolved	○	○	●
Stock would go negative at approval	○	○	●

Two rules cannot be released by anyone: an unresolved blocking doubt, and negative stock.

THE AUDIT CHAIN

Confirm	Order clerk	19:42
Approve	Authorized approver	19:47
Issue	System	19:47

THE DIFF RECORD

The AI draft is stored beside the human's final version, field by field, with a reason code — the same record serves as the evaluation set.

THE WAREHOUSE VIEW

Picking lines only. No prices, no phone numbers, no customer history — a role sees the minimum its task needs.

Verified by test: the clerk's token calling the owner endpoint returns 403 · three roles × two tenants

What we measured — and what we refuse to print.

Runtime results from the extraction module, and a separate latency benchmark. Nothing here is converted into an accuracy claim.

RUN RESULT · 23 AI-ONLY CASES · 24 OCT 2026

23/23	0	0
calls in LLM mode	fallbacks	requests over 4.1 s

WARM-STATE LATENCY · 20 REQUESTS · 23 OCT 2026 · SEPARATE BENCHMARK

2.32 s	2.97 s	3.08 s
p50 median	p90	p95 tail, not a mean

ACCURACY: NOT_COMPUTED

The development inputs have no independent answer key, so we report runtime only — not accuracy.

dev41_ai_only_v1_prod_2026-10-24.json · snapshot 707d45d3 · app/ai@db6f0ce

Runtime evidence, module scope. Acceptance of the full order lifecycle is the next milestone, and it is not claimed here.

WE WILL NOT PRINT

- × An error-rate reduction without a denominator.
- × A fallback result described as AI.
- × “Fully automatic intake” — a person approves every DO.
- × A synthetic order described as a real SME.
- × A price per order.

Sources: evidence/EVIDENCE_INDEX.md · runpod_benchmark_2026-10-23.txt (lines 33–35)

Seven unit gaps. Twelve workflow cases still not run.

CURRENT_WITH_KNOWN_UOM_GAPS

7

known unit-of-measure gaps across 31 output lines

3

alias-injection gaps — “botol”, “pail”: the alias exists in the tenant seed, the adapter does not read it.

4

bulk-kilogram regressions — bulk weight is excluded as if it were a pack size.

WHAT WE FIX NEXT

01 **Add the missing aliases**

and separate bulk kilograms from pack size in the resolver.

02 **Re-run the evaluation**

the 23-case AI-only run, on the fixed AI code, with the same freeze manifest.

03 **Validate the full workflow**

order lifecycle end to end — 12 of the 32 frozen cases are still NOT_RUN.

The honest sentence:

this is module-level evidence, not full order-lifecycle acceptance. A judge who reads the repo will find the same gaps listed there, with the same line count.

SDG 8.2 and 8.3, measured in seconds.

SDG 8
DECENT WORK AND ECONOMIC GROWTH

The impact we claim is economic and operational. We claim no environmental impact, and no effect on jobs or income.

TARGET	8.2 productivity through technological upgrading · 8.3 support for SME productive activity
BASELINE	≈ 210 s median per 5-line order; up to 30 % transcription or stockout discrepancies — a team planning estimate, to be replaced by the pilot
KPI	median seconds from message to a correct, approved DO — including the clarification; plus silent mismatches, price-condition recall and first-use completion
PHASE 2 TARGET	≥ 50 % lower median (≈ 105 s if 210 s validates) · 0 silent mismatches · 100 % of price conditions to a human · ≥ 80 % first-use completion
NOT CLAIMED	no environmental impact claim · no claim on income, jobs or waste

Targets committed before the measurement, not after · source: UN Goal 8, sdgs.un.org/goals/goal8

What runs, what is simulated, what is not measured.

Read this column by column, not row by row — the middle column is what most decks leave out.

RUNS TODAY	SIMULATED — AND LABELLED	NOT MEASURED YET
Five steps end to end on a public link: paste, parse, resolve doubts, approve, DO + stock + log.	Orbyte identity and SSO — demo tokens, not single sign-on.	Tool-side timing and cold start — both scheduled.
Three roles enforced in the backend — the clerk's token gets a 403 on the owner endpoint.	The accounting flag — a synthetic field, not an integration.	A/B timing against a manual baseline with non-developers.
Two tenants on one codebase: switching changes catalogue, thresholds and rules.	Inbound WhatsApp — the clerk pastes; no channel is connected.	First-use completion — the pilot that turns 210 s into a measured number.

HARDEST RISK

One unreleased unit alias turns a correct line into a silent mismatch — and a live model can be cold at 6 pm on a Friday.

THE FALLBACK

The alias and rule path resolves without the model; Local Demo runs without the backend; the backup recording runs without the venue's Wi-Fi.

A capability Orbyte does not yet have, built on its own pattern.

We are not claiming to be part of Orbyte. We are claiming that this is the shape a reusable Orbyte module takes.

ORBYTE PRINCIPLE	HOW KIRAORDER IMPLEMENTS IT
Connected business hub	catalogue, aliases, prices and stock read from one tenant schema
Roles and permissions	three roles × two tenants, enforced server-side
Human approval	consequential actions routed to the role allowed to release them
Activity record	one audit chain per order, with names and timestamps
Reusable across SMEs	a new vertical is configuration, not a fork

WHAT WE DO NOT CLAIM

that we are listed on Orbyte, endorsed by it, connected to WhatsApp, or priced per order.

ONE CODEBASE, TWO TENANTS

Food wholesale at RM 1,500 and hardware cleaning at RM 800: thresholds, catalogue and rules are configuration, not code.

food wholesale · hardware cleaning

THE BUSINESS MODEL

Charged per approved delivery order — an assumption, and no price is printed. A counter of approved DOs exists today.

per approved DO · no price claimed

A wholesaler should not need an ERP to answer one message.

In Malaysia the smallest businesses already run on WhatsApp, a price file and one person’s memory. That is not a gap to be fixed by a bigger system.

It is a gap to be fixed by a system that stops at the right moment, says what it does not know, and knows who does.

Two days after the order, nobody can say who changed the price. We want every small wholesaler to have an order desk that explains itself.

It knows when not to guess — and who to ask.

THREE QUESTIONS WE CAME WITH

01 Is the doubt the right unit to innovate on — or should we pick one kind and go deeper?

02 Are 32 frozen orders evidence, or just a rehearsal?

03 If we had to drop one of the four destinations, which break would cost the least?

PROTOTYPE AND CODE

LIVE DEMO	hozier123.github.io/sdc-2026-team-project
API	kiraorder-api.replit.app/api/v1
GITHUB	github.com/Hozier123/sdc-2026-team-project
BACKUP	recorded pitch + Local Demo (browser-only)

Falcons · five members · Xiamen University Malaysia · SDC Hackathon 2026, Domain 2

Thank you — the next five minutes are yours.